

The Organization of Corruption in Public Procurement

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Abstract

1 Introduction

Public procurement is one of the largest channels through which governments transfer resources to the private sector, and it is widely regarded as a domain where corruption imposes some of its most severe economic and political costs. The existing literature approaches procurement corruption from three directions that never meet. A large detection literature starts with bidding data and tries to infer that something corrupt happened—flagging statistical anomalies in bid clustering, participation patterns, or price dispersion—but these screens cannot identify the actual mechanism: was it bid rotation, specification tailoring, or a kickback arrangement with no bid manipulation at all (Conley & Decarolis, 2016; Harrington, 2008; Imhof et al., 2018; Kawai

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& Nakabayashi, 2022)? A second strand uses corruption revelations from audits or judicial cases as a binary treatment variable, studying what happens when corruption is exposed but never classifying which type of scheme was at work or looking for type-specific signatures in procurement data (Avis et al., 2018; Ferraz & Finan, 2008; Olken, 2007; Zamboni & Litschig, 2018). A third, qualitative literature provides rich taxonomies of how corruption is organized—bid rigging, bribery for contract steering, quality shading, post-award renegotiation—but these typologies remain descriptive and are not systematically tested against actual procurement outcomes (Konov & Gorbacheva, 2023; OECD, 2016; Søreide, 2002). As a result, we still lack a systematic understanding of how corruption in public procurement is actually organized—which specific schemes are used at scale, under what conditions they arise, and how they manifest in observable bidding and contract data.

In this paper, we provide the first large-scale empirical study of the organizational mechanics of corruption in public procurement. We assemble a comprehensive dataset that links the full universe of public contracts and bid histories to detailed judicial case files involving procurement-related crimes. By integrating these data, we can observe not only which contracts were ultimately exposed as corrupt, but also the structure of the auctions in which they originated, the firms that participated in them, the evolution of contract values through amendments, and the broader network of relationships among firms and public officials. Using large language models to systematically code judicial narratives, we identify a set of recurrent scheme types and link each to its distinctive procurement signature. This unified view allows us to move beyond the identification of suspicious outcomes and instead charac-

terize the specific schemes through which corruption is carried out—and to examine how these mechanisms vary across institutional environments and political conditions.

Our analysis yields three central findings. First, by leveraging detailed judicial case files, we uncover the operational mechanics through which corrupt contracts were executed—how bribes were arranged, when they were paid, how bureaucrats manipulated bidding rules, and how favored firms were steered toward victory. Using large language models to code these narratives at scale, we identify a typology of recurrent scheme types that would be exceedingly difficult to classify with traditional manual coding or keyword methods. Second, when these scheme-level classifications are linked to the universe of procurement data, clear quantitative patterns emerge: each scheme has a distinct bidding signature and ex post contract trajectory. For example, schemes involving pre-award bribe payments and bid tailoring feature almost no genuine competition and minimal price dispersion, whereas schemes involving post-award kickbacks exhibit aggressive underbidding followed by rapid contract amendments or systematic quality shading. Third, corruption is embedded in a durable network structure: the same firms and intermediaries reappear across multiple schemes, sometimes replicating the same *modus operandi* across different contracting authorities.

This study is feasible because of a rare combination of data available in Brazil’s state of São Paulo. We draw on more than 12,000 full-text judicial decisions in *improbidade administrativa* cases from the São Paulo State Court of Justice (TJSP), which describe in detail how corruption was organized—the specific scheme, the actors, and the mechanics. We link these cases to bid-

level municipal procurement records from the São Paulo State Audit Court (TCE-SP), covering 644 municipalities with firm identifiers, bid values, and full contract histories. A chain of linkable identifiers—procurement process numbers extracted from judicial texts matched to administrative records—connects the narrative evidence to the quantitative data. Large language models allow us to classify scheme types from judicial narratives at scale, replacing what would otherwise require thousands of hours of expert legal coding. Most countries lack all three ingredients: detailed judicial narratives, granular procurement data, and linkable identifiers. Brazil’s combination of these elements creates a unique empirical opportunity.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the institutional setting, including Brazilian municipal procurement rules, the role of the *Ministério Público* in corruption enforcement, and procurement oversight by the *Tribunal de Contas*. Section 4 presents the data sources and the linking procedure. Section 5 details the LLM-based scheme classification and validation. Section 6 lays out the empirical strategy. Sections 7 and 8 present the main results and network analysis. Section 9 reports robustness checks. Section 10 discusses implications, and Section 11 concludes.

2 Related Literature

This paper connects three literatures that have developed largely in isolation: empirical screening for procurement anomalies, the study of corruption through audit and judicial revelations, and qualitative typologies of how cor-

ruption is organized. We also draw on recent work using large language models for text classification in social science and on the empirical literature on Brazilian procurement institutions.

2.1 Detection and screening

A large literature develops statistical screens to detect collusion and corruption in procurement data. Harrington (2008) lays out the theoretical foundations for cartel screening, identifying patterns in bid levels, bid dispersion, and winner identity that coordinated behavior should produce. Imhof et al. (2018) operationalize these ideas, showing that the coefficient of variation of bids, bid-level regressions, and screens for cover bidding can discriminate between competitive and collusive auctions. Kawai and Nakabayashi (2022) exploit rebid data from Japanese procurement to identify over 1,000 firms engaged in non-competitive conduct, while Conley and Decarolis (2016) develop statistical tests for coordinated entry and bidding, validated against Italian court cases.

On the corruption side, Palguta and Pertold (2017) document bunching of contract values just below discretionary thresholds in Czech procurement, with a threefold increase in awards to anonymous firms below the cutoff—a signature of contract splitting to avoid oversight. Decarolis and Giorgiantonio (2022) use machine learning to predict corruption risk from Italian tender characteristics, finding that combinations of red flags substantially outperform individual indicators. Harrington and Imhof (2022) connect modern data methods to cartel detection, establishing what machine learning can

and cannot identify from bidding data.

The limitation common to this strand is that screens detect *anomalies* but cannot identify the underlying *mechanism*. A cluster of low-dispersion bids could reflect bid rotation by a cartel, cover bidding by shell companies, or specification tailoring that discourages genuine entry. Without knowing which scheme produced the pattern, it is difficult to design targeted countermeasures or to assess the welfare consequences. Our contribution is to resolve this ambiguity by linking screen-detectable anomalies to judicially identified scheme types.

2.2 Audits, enforcement, and revealed corruption

A second strand uses corruption revelations—typically from random audits or judicial proceedings—as a treatment variable, studying the consequences of exposure rather than the mechanics of the underlying conduct. The pioneering work by Ferraz and Finan (2008) shows that randomly timed audit reports revealing procurement fraud in Brazilian municipalities reduce incumbents’ reelection probability. Ferraz and Finan (2011) extend this to show that reelection incentives discipline corruption: term-limited mayors steal 27 percent more than reelection-eligible ones. Avis et al. (2018) estimate that the deterrence effect of audits reduces corruption by approximately 8 percent in subsequently audited municipalities. Olken (2007) finds that increasing the probability of a government audit from 4 to 100 percent reduced missing expenditures by 8 percentage points in Indonesian road projects. Zamboni and Litschig (2018) show that higher perceived audit probability

lowers procurement fraud in Brazil, especially where legal consequences are credible.

These papers treat corruption as a binary variable: either an irregularity was found or it was not. They do not classify *which type* of scheme was at work—whether the fraud involved bid rigging, contract splitting, kickbacks, or specification tailoring—nor do they ask whether different scheme types produce different procurement signatures. We build on this tradition by using judicial revelations not as a binary treatment but as a source of detailed scheme-level classifications that can be linked back to procurement data.

2.3 Corruption typologies and cartel organization

A third literature provides rich qualitative taxonomies of how procurement corruption is organized. Søreide (2002) offers an early comprehensive typology distinguishing bribery, collusive bid arrangements, and administrative frauds across procurement stages. Konov and Gorbacheva (2023) catalog specific manipulation tactics—specification tailoring, phantom bids, contract splitting, fake competition—drawing on case studies across jurisdictions. The OECD has published authoritative taxonomies of bid-rigging forms and detection checklists (OECD, 2009, 2016). In Brazil, CADE’s *Guia de Combate a Cartéis em Licitação* describes cover bidding, bid suppression, rotation, and market division based on two decades of enforcement experience, while the *Tribunal de Contas da União* (TCU) publishes a framework for fraud detection that includes quantitative screening indicators.

Campos et al. (2021) provide the closest precedent to our approach: a

detailed study of the Odebrecht case documenting how bribes were arranged to tailor auction terms and obtain favorable renegotiations. However, this remains a single-firm case study. Our contribution is to systematically classify scheme types across hundreds of judicial cases and link the resulting typology to quantitative procurement outcomes, providing the empirical test that this descriptive literature has lacked.

2.4 Text as data and LLM classification

Our use of large language models to code judicial narratives at scale connects to a rapidly growing literature on text-as-data methods in economics and political science. Gentzkow et al. (2019) provide the foundational survey of text analysis in economics, while Ash and Hansen (2024) offer a field-level overview of how LLMs are transforming text-based research. Ash et al. (2023) demonstrate that full-text analysis of court opinions can reveal ideological shifts in the judiciary, establishing that legal documents contain rich extractable information beyond what keyword methods capture.

Several recent papers validate the use of LLMs as substitutes for human coders. Gilardi et al. (2023) show that zero-shot ChatGPT matches or exceeds crowd-worker accuracy on political text annotation at substantially lower cost. Heseltine and von Hohenberg (2024) benchmark GPT-4 against expert coders and propose hybrid workflows for maintaining validity. Törnberg (2024) develops practical standards for reproducibility and reliability in LLM-based annotation. We apply these methods to a new domain —extracting structured scheme classifications from Brazilian judicial

narratives—and validate against expert legal coding.

2.5 Procurement institutions in Brazil

Our institutional setting builds on a substantial literature studying procurement and corruption in Brazil. Pereira and Tonin (2023) provide an overview of the anti-corruption framework, discussing the interplay between Lei 8.666/1993, the *Ministério Público*, audit courts, and the 2021 reforms. Litschig and Zamboni (2024) show that municipalities with a local judiciary have approximately 10 percent less resource diversion, suggesting that the proximity of enforcers shapes corruption opportunities. Romero (2025) finds that more capable bureaucracies paradoxically increase the likelihood of connected firms winning through less competitive processes, implying that scheme sophistication co-evolves with administrative capacity. Colonnelli and Prem (2022) show that audits trigger firm entry in government-dependent sectors, while Boas et al. (2014) document that firms donating to winning political campaigns receive at least 14 times the value of their contributions in subsequent procurement contracts.

On procurement design, Carril (2021) studies the effects of scrutiny thresholds in U.S. defense procurement, finding that added scrutiny above thresholds distorts awards and worsens ex post performance—a finding directly relevant to our analysis of threshold manipulation in Brazil. Decarolis et al. (2025) show that discretionary auctions are linked to corruption only when competition is restricted, highlighting the interaction between auction format and corruption opportunity that our scheme-level analysis can illuminate.

3 Institutional Background

Our empirical setting is municipal procurement in the state of São Paulo, Brazil. This section describes the procurement rules that create the opportunities our scheme typology exploits, the legal framework through which corruption is prosecuted (generating our judicial data), the audit institutions that produce procurement records, and the competition authority whose cartel cases provide additional revealed corruption.

3.1 Brazilian municipal procurement

For the bulk of our sample period, municipal procurement was governed by Lei 8.666/1993, which established five procurement modalities with value thresholds determining which modality applies. The two most relevant for our analysis are *convite* (invitation to at least three firms) and *concorrência* (open to all qualified bidders). For engineering works, *convite* applied up to R\$150,000, *tomada de preços* up to R\$1,500,000, and *concorrência* above that threshold; for goods and services, the limits were R\$80,000 and R\$650,000, respectively.¹ Below these thresholds, the law also permitted direct contracting (*dispensa de licitação*) for small purchases: up to R\$15,000 for works and R\$8,000 for other purchases (Art. 24, I–II). These low-value thresholds are the margins most commonly exploited through contract splitting (*fracionamento*), which Art. 23 §5 explicitly prohibits.

Lei 10.520/2002 introduced the *pregão* (reverse auction), an open descending-price format that became the most common modality for goods and standard

¹Values from Lei 8.666/1993, Art. 23, as updated by Decreto 9.412/2018.

services. The electronic version (*pregão eletrônico*) was widely adopted by municipalities and is well represented in our data.

In 2021, Lei 14.133 replaced the entire procurement framework, substantially raising the direct-contracting thresholds to R\$100,000 for works and R\$50,000 for goods (Art. 75, I–II). A transition period allowed municipalities to operate under either the old or new regime until December 2023. Most of our sample period predates this reform, but the threshold increase provides a natural experiment for testing whether corruption patterns shift in response to regulatory changes.

Contract amendments (*aditivos*) are permitted under Art. 65 §1 of Lei 8.666/1993, up to 25 percent of the original value for goods and services and 50 percent for construction renovations. These statutory limits are important for our analysis of post-award corruption: schemes involving overpricing through amendments (*superfaturamento*) and the manipulation of unit quantities (*jogo de planilha*) operate within and against these constraints.

3.2 Improbidade administrativa and the Ministério Público

Our judicial data consists of *sentenças* (first-instance decisions) in *ações de improbidade administrativa*, the primary civil mechanism for prosecuting corruption in public administration. Lei 8.429/1992 establishes three categories of administrative improbity: illicit enrichment (Art. 9), damage to the public treasury (Art. 10), and violation of administrative principles (Art. 11). Sanctions include loss of public office, suspension of political rights for up to 14 years, fines, and prohibition from government contracting.

The *Ministério Público* (MP) has near-exclusive standing to bring improbidade actions.² Investigations typically follow a pipeline: a *notícia de fato* (initial report, processed within 120 days) leads to an *inquérito civil* (civil investigation, nominally one year but frequently extended), which may result in an *ação civil pública* filed before the state court. In São Paulo, the state-level MP (MP-SP) brings cases before the *Tribunal de Justiça do Estado de São Paulo* (TJSP), the largest state court in Brazil.

Two features of this institutional setting are important for our research design. First, the typical lag from the corrupt act to a first-instance *sentença* is 7–8 years, meaning that our judicial sample primarily captures corruption that occurred in earlier periods. Second, the 2021 reform eliminated liability for negligent (*culposa*) improbidade, requiring proof of intentional (*dolosa*) conduct. This raises the evidentiary bar but also means that the cases in our sample that result in conviction tend to contain particularly detailed descriptions of corrupt mechanics, which is advantageous for our scheme classification exercise.

3.3 The Tribunal de Contas and procurement oversight

The *Tribunal de Contas do Estado de São Paulo* (TCE-SP) is the external audit body responsible for overseeing municipal public accounts in 644 non-capital municipalities in São Paulo state.³ TCE-SP performs annual account reviews and can audit specific procurement processes, impose fines, and sanction officials.

²After the 2021 reform (Lei 14.230/2021), only the MP can initiate improbidade proceedings; previously, the affected public entity could also file suit.

³The capital has its own audit court (TCM-SP).

Critically for our study, TCE-SP operates the *Audesp* (Auditoria Eletrônica de Órgãos Públicos) system, through which all municipalities must upload detailed procurement and contract data. The Audesp system records bidding process identifiers, modality, estimated and actual values, participating firms (by CNPJ), bid amounts, contract values, and amendment histories. This is the source of our procurement data and provides the quantitative counterpart to the qualitative judicial narratives.

TCE-SP findings can trigger MP investigations but are legally distinct from judicial proceedings: a TCE-SP irregularity finding does not constitute a judicial determination of improbidade. In practice, however, TCE-SP reports appear as evidence in many of the improbidade cases in our sample, providing a link between the audit and judicial systems.

3.4 CADE and cartel enforcement

The *Conselho Administrativo de Defesa Econômica* (CADE) is Brazil’s competition authority, with jurisdiction over anticompetitive agreements including bid rigging in public procurement. Under Lei 12.529/2011, agreements between competitors to fix prices, divide markets, or rig bids are administrative infractions subject to fines of 0.1 to 20 percent of gross revenue.

CADE’s leniency program (Arts. 86–87) provides partial or full immunity to the first confessor, generating detailed descriptions of cartel mechanics that serve as an additional source of revealed corruption in our analysis. Several major CADE cases involve São Paulo procurement: the school meals cartel (*merenda escolar*), operating across approximately 30 municipalities

for over 20 years; the trains and metro cartel (*trens e metrô*s), involving 19 firms in maintenance contracts; and infrastructure cartels linked to the *Rodoanel* highway program. These cases provide rich institutional detail on how cartels organize bid rotation, cover bidding, and market division in Brazilian procurement.

The same conduct can be prosecuted in parallel across multiple legal channels: as an administrative infraction by CADE, a criminal offense under Lei 8.137/1990 and Lei 8.666/1993, an improbidade action by the MP, and a corporate liability proceeding under Lei 12.846/2013. This multiplicity of enforcement channels means that the same corrupt arrangement may appear in our data through different institutional lenses, which we exploit for cross-validation.

4 Data

Our analysis requires two main data sources—judicial case files describing how corruption was organized and procurement records describing how auctions and contracts unfolded—plus a linking procedure that connects them.

4.1 Court cases

We obtain the universe of first-instance decisions (*sentenças*) in improbidade administrativa cases from the TJSP electronic system. The full corpus contains over 12,000 cases. We focus on cases with procurement-related content, identified through keyword filtering and subject-matter classification.

Each *sentença* is a full-text judicial document, typically 10–40 pages,

that describes the facts of the case as established by the court: who was involved, what they did, how the corrupt arrangement was organized, and what evidence supported the finding. For cases resulting in conviction, these narratives are particularly detailed, as the judge must establish the elements of improbidade—intentional conduct (*dolo*), damage to the public treasury or violation of administrative principles, and the causal link between the two. This level of detail is what allows us to classify scheme types: the *sentença* typically describes whether the corruption involved contract splitting, bid steering, shell companies, kickbacks, or other mechanisms.

4.2 Procurement records

We use two complementary datasets from the TCE-SP AudeSP system covering 644 municipalities in São Paulo state:

Licitações (bid-level data) record each bidding process with the participating firms, their bid values, qualification status, and outcome (winner, classified, disqualified, no-show). Variables include the procurement process identifier, modality, object description, estimated value, and the date of the auction. Each row represents a bidder–item pair within a procurement event.

Ajustes (contract-level data) record the resulting contracts: the winning firm’s CNPJ, contracted value, start and end dates, and any subsequent amendments (*aditivos*) with their values. Contracts are linked to the originating bidding process through a procurement code, allowing us to trace the full lifecycle from auction to execution.

The combination of bid-level and contract-level data is essential for our

analysis. The bid-level data allow us to compute auction-level statistics — number of bidders, bid dispersion, winning margins, participation patterns — that form the “bidding signatures” in our scheme comparison. The contract-level data allow us to track post-award outcomes—amendment frequency and magnitude, cost overruns—that distinguish pre-award from post-award corruption schemes.

4.3 Linking court cases to procurement data

The key empirical challenge is connecting the two data sources. Judicial decisions in improbidade cases frequently reference specific procurement processes by number (e.g., “Pregão 26/2010” or “Carta Convite 03/2015”). We extract these references using a set of regular expression patterns applied to the full text of each *sentença*, capturing modality keywords followed by process numbers and years.

Three extraction patterns are applied in order of confidence: (i) a modality keyword (*pregão, concorrência, tomada de preços, convite, dispensa*) followed by a number and year (high confidence); (ii) a generic number–year pattern (e.g., “nº 26/2010”) appearing within 300 characters of a procurement keyword (medium confidence); (iii) an administrative process (*processo administrativo*) prefix with procurement context. We validate extraction quality against independent LLM-based extraction on two random samples of 50 cases each, achieving recall of 96–100 percent against true procurement references.

Extracted references are matched to TCE-SP *ajustes* on the combination

of municipality (derived from the court’s *comarca*), process number, and year. Municipality names are normalized by stripping accents and standardizing formatting, achieving resolution for 98.4 percent of *comarcas*. Through the procurement code in the *ajustes* data, matched cases are further linked to bid-level records in the *licitações* data, yielding the complete auction–contract–judicial chain.

4.4 Descriptive statistics

5 Scheme Classification

The central methodological innovation of this paper is the use of large language models to systematically classify corruption schemes from judicial narratives. This section describes the coding protocol, the resulting typology, and how we validate the classifications.

5.1 LLM-based coding of judicial narratives

Each *sentença* in our sample is processed by a large language model tasked with extracting structured information about the corruption scheme described in the case. The prompt instructs the model to identify: (i) the type(s) of corruption scheme (*fracionamento*, bid steering, shell companies, cartel coordination, kickbacks, overpricing, post-award manipulation, or combinations thereof); (ii) the actors involved and their roles (initiator, beneficiary, cover bidder, intermediary); (iii) the specific procurement processes referenced; (iv) the mechanics of the scheme (how bids were coordinated, how bribes

were calculated, how oversight was evaded); and (v) the evidence basis (documentary, testimonial, expert, financial forensic).

We use a structured output format that enforces consistent categorization while allowing the model to capture case-specific detail in free-text fields. The prompt is designed following best practices for LLM-based annotation in social science (Törnberg, 2024): it includes explicit definitions of each scheme type grounded in Brazilian legal categories and institutional guides, provides worked examples from pilot coding, and instructs the model to flag ambiguous cases rather than force classification.

5.2 Typology of corruption schemes

The classification produces a typology of corruption schemes organized along two dimensions: the *stage* at which the manipulation occurs (pre-award vs. post-award) and the *organizational structure* of the corrupt arrangement (bilateral vs. network).

Pre-award schemes manipulate the bidding process itself:

- *Fracionamento* (contract splitting): Splitting a single procurement need into multiple contracts below the threshold for competitive bidding, enabling direct contracting with a favored firm.
- *Direcionamento* (bid steering / specification rigging): Tailoring edital specifications, technical requirements, or qualification criteria to ensure a predetermined winner, often combined with undue disqualification of competitors.

- *Empresas de fachada* (shell/phantom companies): Creating or deploying firms with no genuine commercial activity to submit cover bids, fill minimum-bidder requirements for *carta-convite*, or simulate competitive processes.
- *Cartel / conluio* (bid rigging): Coordinated bidding among genuinely operating firms through bid rotation, market division, cover bidding, or bid suppression, often facilitated by intermediaries or public officials.

Post-award schemes extract rents during contract execution:

- *Superfaturamento* (overpricing): Inflating contract prices above market rates, either at the contracting stage (*sobrepreço*) or through the manipulation of unit quantities in amendments (*jogo de planilha*).
- *Propina / kickbacks*: Direct payments from contractors to officials, calculated as a percentage of contract value or periodic measurements, in exchange for contract award, favorable amendments, or lax oversight.
- *Phantom delivery*: Payments for goods never delivered or services never performed, supported by fraudulent invoices and complicit contract supervisors.

These categories are not mutually exclusive: a single case may involve contract splitting to avoid competitive bidding *and* overpricing of the resulting contracts, or cartel coordination in bidding *and* kickbacks to officials who facilitated the arrangement. Our classification allows multiple scheme labels per case, capturing these combinations.

5.3 Validation

We validate the LLM classifications along three dimensions.

Inter-rater reliability. A random subsample of cases is independently coded by research assistants with legal training. We compute Cohen’s kappa for each scheme category and report the confusion matrix between LLM and human classifications. Following Gilardi et al. (2023), we expect LLM accuracy to be comparable to inter-coder agreement among human annotators.

Prompt robustness. We assess sensitivity to prompt formulation by varying the instructions (e.g., providing definitions vs. examples, changing the order of categories, using Portuguese vs. English prompts) and measuring classification stability across prompt variants.

External consistency. For the subset of cases that overlap with CADE cartel decisions or Lava Jato denúncias, we verify that the LLM classification is consistent with the independently established facts in those proceedings. We also compare our typology to existing taxonomies (Konov & Gorbacheva, 2023; Søreide, 2002) to ensure conceptual coverage.

6 Empirical Strategy

Our empirical analysis proceeds in three parts, corresponding to the three central findings previewed in the introduction: (i) we compare bidding signatures across scheme types, (ii) we compare contract execution trajectories, and (iii) we analyze the network structure of corrupt arrangements. This section describes the outcome variables, sources of variation, and specifications for each part, as well as the identification concerns common to all three.

6.1 Bidding signatures by scheme type

The first part of the analysis asks whether different corruption schemes produce distinct patterns in bidding data. For each procurement process in the TCE-SP data, we compute a set of auction-level statistics motivated by the screening literature and our case evidence:

- *Number of bidders*: the count of firms submitting bids.
- *Single-bidder indicator*: whether only one firm bid.
- *Bid dispersion*: the coefficient of variation of bid values, capturing the degree of genuine price competition.
- *Winning margin*: the gap between the winning bid and the second-lowest bid, as a share of the winning bid.
- *Price ratio*: the winning bid divided by the estimated value, measuring the degree of discount (or premium) relative to the procuring entity's benchmark.
- *Threshold proximity*: the distance of the contract value from the nearest statutory threshold, capturing bunching behavior associated with contract splitting.

Our core specification compares these outcomes across scheme types:

$$Y_{ijmt} = \sum_{s \in \mathcal{S}} \beta_s \cdot \mathbf{1}[\text{Scheme}_i = s] + \mathbf{X}_i' \gamma + \alpha_m + \delta_t + \theta_j + \varepsilon_{ijmt}, \quad (1)$$

where Y_{ijmt} is the bidding outcome for procurement process i in object category j , municipality m , and year t ; $\mathbf{1}[\text{Scheme}_i = s]$ indicates whether the process is linked to a judicial case classified as scheme type s ; $\mathbf{X}_i$ is a vector of process-level controls (estimated value, modality); α_m , δ_t , and θ_j are municipality, year, and object-category fixed effects. The coefficients β_s measure the difference in bidding outcomes between processes linked to each scheme type and the omitted category of non-corrupt processes.

Municipality fixed effects absorb time-invariant differences in local market structure, administrative capacity, and political environment. Year fixed effects absorb aggregate trends in procurement practices. Object-category fixed effects absorb differences in market thickness and competitive conditions across procurement sectors.

For the specific test of threshold bunching associated with *fraccionamiento*, we supplement the regression analysis with density discontinuity tests at each statutory threshold. Following Palguta and Pertold (2017), we estimate the density of contract values separately for municipalities with revealed *fraccionamiento* cases and matched controls, testing whether excess mass below the threshold is concentrated in the corrupt sample.

6.2 Contract execution trajectories

The second part examines post-award outcomes. For each contract in the TCE-SP *ajustes* data, we compute:

- *Amendment indicator*: whether the contract received any amendment.
- *Amendment count*: the number of amendments.

- *Amendment magnitude*: the cumulative amendment value as a share of the original contract value.

We estimate a specification parallel to equation (1), with contract-level execution outcomes as the dependent variable and the same fixed-effects structure.

The key prediction, drawn from the case evidence and the renegotiation literature (Bajari et al., 2014; Iimi, 2013; Jung et al., 2019), is that pre-award and post-award corruption schemes have *opposite* bidding-amendment profiles. Pre-award schemes (specification rigging, bid steering) produce winning prices at or above the estimated value with few subsequent amendments, because the rents are extracted at the award stage. Post-award schemes (*jogo de planilha*, kickbacks tied to contract measurements) produce aggressive underbidding—firms win with large discounts—followed by substantial amendments that restore or exceed the original value. We test this prediction by jointly modeling the price ratio and amendment magnitude, interacting scheme-type indicators with both outcomes.

6.3 Network analysis

The third part analyzes the structure of relationships among firms and officials involved in corrupt procurement. We construct a bipartite firm–process network from the bid-level data, where firms are linked to the procurement processes in which they participate. Within this network, we identify subgraphs associated with judicially revealed corruption and compare their topology to subgraphs drawn from the non-corrupt portion of the network.

Our case evidence suggests that corrupt networks exhibit a hub-and-spoke structure, with intermediary firms or officials occupying central positions. We test this by computing standard network statistics—degree distribution, betweenness centrality, clustering coefficient—for corruption-linked and control subnetworks, and testing for differences using permutation inference.

We also examine network durability: whether firms flagged in judicial cases during one mayoral administration continue winning contracts at elevated rates under the subsequent administration. Using mayoral turnover as a natural break in the political environment, we estimate a difference-in-differences specification comparing the win rates of flagged versus non-flagged firms before and after administration changes.

6.4 Identification concerns

Our analysis faces three main identification challenges.

Selection of revealed corruption. Judicial cases represent a selected sample of all corruption: only schemes that were detected, investigated, prosecuted, and adjudicated appear in our data. If detection probability varies with scheme type—for instance, if crude contract splitting is easier to detect than sophisticated bid rigging—then the distribution of scheme types in our judicial sample may differ from the true population distribution. We address this concern in two ways. First, for the bidding-signature analysis, selection on detection does not bias the *comparison across scheme types*, which is our primary object of interest, unless detection probability is differentially correlated with bidding outcomes conditional on scheme type. Second, we

examine whether our results are robust to restricting the sample to cases triggered by different detection channels (TCE-SP audits, police investigations, citizen reports), which plausibly have different selection properties.

Measurement error in scheme classification. The LLM-based classification may assign incorrect scheme labels. Classical measurement error in a categorical treatment variable attenuates differences across categories, biasing our scheme-type coefficients toward zero. Our validation exercise (Section 5.3) quantifies the error rate and allows us to assess the degree of attenuation. For robustness, we also estimate specifications using only cases where the LLM classification is confirmed by human coders.

Linking incompleteness. Not all judicial cases contain extractable procurement process numbers, and not all extracted numbers match to TCE-SP records. If the probability of a successful match varies with case characteristics, the linked sample may be non-representative. We report match rates by case type and municipality, and compare observable characteristics of matched and unmatched cases to assess the severity of this concern.

7 Results

7.1 Scheme prevalence and characteristics

7.2 Bidding signatures by scheme type

7.3 Contract execution trajectories

8 Network Analysis

9 Robustness

10 Discussion

11 Conclusion

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A Additional Tables and Figures

B LLM Prompt and Coding Protocol